

15.2 Configure IP Static Routes

Configure IP Static Routes

IPv4 Next-Hop Static Route

In a next-hop static route, only the next-hop IP address is specified. The exit interface is derived from the next hop. For example, three next-hop IPv4 static routes are configured on R1 using the IP address of the next hop, R2.

```
R1(config)# ip route 172.16.1.0 255.255.255.0 172.16.2.2
```

```
R1(config)# ip route 192.168.1.0 255.255.255.0 172.16.2.2
```

```
R1(config)# ip route 192.168.2.0 255.255.255.0 172.16.2.2
```

The resulting routing table entries on R1:

```
R1# show ip route | begin Gateway
Gateway of last resort is not set
    172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
S       172.16.1.0/24 [1/0] via 172.16.2.2
C       172.16.2.0/24 is directly connected, Serial0/1/0
L       172.16.2.1/32 is directly connected, Serial0/1/0
C       172.16.3.0/24 is directly connected, GigabitEthernet0/0/0
L       172.16.3.1/32 is directly connected, GigabitEthernet0/0/0
S       192.168.1.0/24 [1/0] via 172.16.2.2
S       192.168.2.0/24 [1/0] via 172.16.2.2
```

Configure IP Static Routes

IPv6 Next-Hop Static Route

The commands to configure R1 with the IPv6 static routes to the three remote networks are as follows:

```
R1(config)# ipv6 unicast-routing
```

```
R1(config)# ipv6 route 2001:db8:acad:1::/64  
2001:db8:acad:2::2
```

```
R1(config)# ipv6 route 2001:db8:cafe:1::/64  
2001:db8:acad:2::2
```

```
R1(config)# ipv6 route 2001:db8:cafe:2::/64  
2001:db8:acad:2::2
```

The routing table for R1 now has routes to the three remote IPv6 networks.

```
R1# show ipv6 route  
IPv6 Routing Table - default - 8 entries  
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route  
B - BGP, R - RIP, H - NHRP, I1 - ISIS L1  
I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP  
EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination  
NDR - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter  
OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1  
ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations  
ld - LISP dyn-eid, la - LISP away, le - LISP extranet-policy  
a - Application  
S 2001:DB8:ACAD:1::/64 [1/0]  
   via 2001:DB8:ACAD:2::2  
C 2001:DB8:ACAD:2::/64 [0/0]  
   via Serial0/1/0, directly connected  
L 2001:DB8:ACAD:2::1/128 [0/0]  
   via Serial0/1/0, receive  
C 2001:DB8:ACAD:3::/64 [0/0]  
   via GigabitEthernet0/0/0, directly connected  
L 2001:DB8:ACAD:3::1/128 [0/0]  
   via GigabitEthernet0/0/0, receive  
S 2001:DB8:CAFE:1::/64 [1/0]  
   via 2001:DB8:ACAD:2::2  
S 2001:DB8:CAFE:2::/64 [1/0]  
   via 2001:DB8:ACAD:2::2  
L FF00::/8 [0/0]  
   via Null0, receive
```

IPv4 Directly Connected Static Route

When configuring a static route, another option is to use the exit interface to specify the next-hop address. Three directly connected IPv4 static routes are configured on R1 using the exit interface.

Note: Using a next-hop address is generally recommended. Directly connected static routes should only be used with point-to-point serial interfaces.

```
R1(config)# ip route 172.16.1.0 255.255.255.0 s0/1/0
```

```
R1(config)# ip route 192.168.1.0 255.255.255.0 s0/1/0
```

```
R1(config)# ip route 192.168.2.0 255.255.255.0 s0/1/0
```

```
R1# show ip route | begin Gateway
Gateway of last resort is not set
    172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
S       172.16.1.0/24 is directly connected, Serial0/1/0
C       172.16.2.0/24 is directly connected, Serial0/1/0
L       172.16.2.1/32 is directly connected, Serial0/1/0
C       172.16.3.0/24 is directly connected, GigabitEthernet0/0/0
L       172.16.3.1/32 is directly connected, GigabitEthernet0/0/0
S       192.168.1.0/24 is directly connected, Serial0/1/0
S       192.168.2.0/24 is directly connected, Serial0/1/0
```

Configure IP Static Routes

IPv6 Directly Connected Static Route

In the example, three directly connected IPv6 static routes are configured on R1 using the exit interface.

Note: Using a next-hop address is generally recommended. Directly connected static routes should only be used with point-to-point serial interfaces.

```
R1(config)# ipv6 route 2001:db8:acad:1::/64  
s0/1/0
```

```
R1(config)# ipv6 route 2001:db8:cafe:1::/64  
s0/1/0
```

```
R1(config)# ipv6 route 2001:db8:cafe:2::/64  
s0/1/0
```

```
R1# show ipv6 route
IPv6 Routing Table - default - 8 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, RL - RPL, O - OSPF Intra, OI - OSPF Inter
       OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1
       ON2 - OSPF NSSA ext 2, la - LISP alt, lr - LISP site-registrations
       ld - LISP dyn-eid, lA - LISP away, le - LISP extranet-policy
       a - Application
S 2001:DB8:ACAD:1::/64 [1/0]
  via Serial0/1/0, directly connected
C 2001:DB8:ACAD:2::/64 [0/0]
  via Serial0/1/0, directly connected
L 2001:DB8:ACAD:2::1/128 [0/0]
  via Serial0/1/0, receive
C 2001:DB8:ACAD:3::/64 [0/0]
  via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:3::1/128 [0/0]
  via GigabitEthernet0/0/0, receive
S 2001:DB8:CAFE:1::/64 [1/0]
  via Serial0/1/0, directly connected
S 2001:DB8:CAFE:2::/64 [1/0]
  via Serial0/1/0, directly connected
L FF00::/8 [0/0]
  via Null0, receive
IPv6 Routing Table - default - 8 entries
R1#
```

Configure IP Static Routes

IPv4 Fully Specified Static Route

- In a fully specified static route, both the exit interface and the next-hop IP address are specified. This form of static route is used when the exit interface is a multi-access interface and it is necessary to explicitly identify the next hop. The next hop must be directly connected to the specified exit interface. Using an exit interface is optional, however it is necessary to use a next-hop address.
- It is recommended that when the exit interface is an Ethernet network, that the static route includes a next-hop address. You can also use a fully specified static route that includes both the exit interface and the next-hop address.

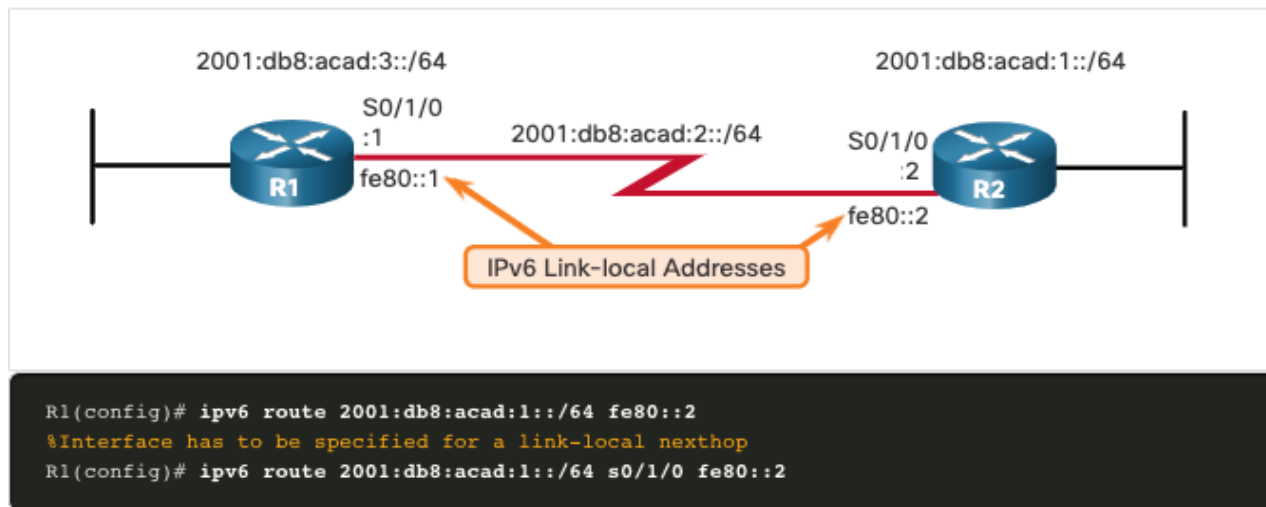
```
R1(config)# ip route 172.16.1.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2
R1(config)# ip route 192.168.1.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2
R1(config)# ip route 192.168.2.0 255.255.255.0 GigabitEthernet 0/0/1 172.16.2.2
```

```
R1# show ip route | begin Gateway
Gateway of last resort is not set
    172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
S       172.16.1.0/24 [1/0] via 172.16.2.2, GigabitEthernet0/0/1
C       172.16.2.0/24 is directly connected, GigabitEthernet0/0/1
L       172.16.2.1/32 is directly connected, GigabitEthernet0/0/1
C       172.16.3.0/24 is directly connected, GigabitEthernet0/0/0
L       172.16.3.1/32 is directly connected, GigabitEthernet0/0/0
S      192.168.1.0/24 [1/0] via 172.16.2.2, GigabitEthernet0/0/1
S      192.168.2.0/24 [1/0] via 172.16.2.2, GigabitEthernet0/0/1
```

Configure IP Static Routes

IPv6 Fully Specified Static Route

In a fully specified static route, both the exit interface and the next-hop IPv6 address are specified. There is a situation in IPv6 when a fully specified static route must be used. If the IPv6 static route uses an IPv6 link-local address as the next-hop address, use a fully specified static route. The figure shows an example of a fully specified IPv6 static route using an IPv6 link-local address as the next-hop address.



IPv6 Fully Specified Static Route (Cont.)

The reason a fully specified static route must be used is because IPv6 link-local addresses are not contained in the IPv6 routing table. Link-local addresses are only unique on a given link or network. The next-hop link-local address may be a valid address on multiple networks connected to the router. Therefore, it is necessary that the exit interface be included.

The following example shows the IPv6 routing table entry for this route. Notice that both the next-hop link-local address and the exit interface are included.

```
R1# show ipv6 route static | begin 2001:db8:acad:1::/64
S    2001:DB8:ACAD:1::/64 [1/0]
    via FE80::2, Serial0/1/0
```


Configure IP Static Routes

Verify a Static Route

Along with **show ip route**, **show ipv6 route**, **ping** and **tracert**, other useful commands to verify static routes include the following:

- **show ip route static**
- **show ip route *network***
- **show running-config | section ip route**

Replace **ip** with **ipv6** for the IPv6 versions of the command.